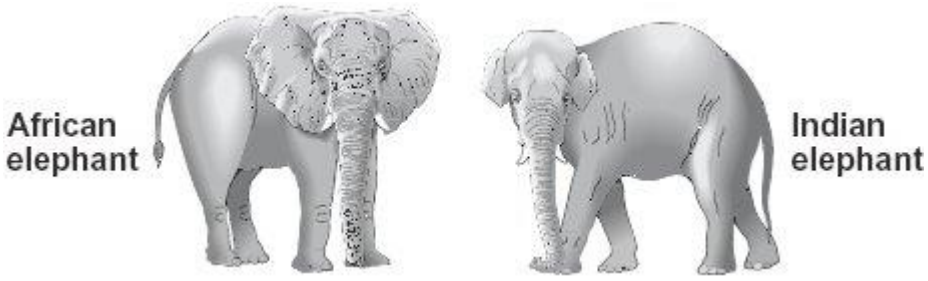
Q1.

(a) Elephants keep cool by losing heat from their ears.



Predict which elephant can lose more heat from its ears.

..... elephant

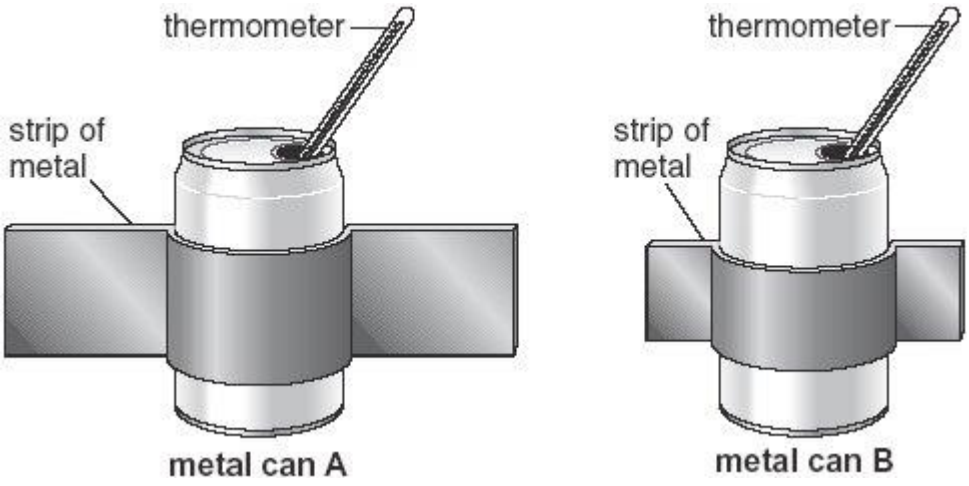
Give the reason for your answer.

.....

.....

1 mark

(b) Ben filled two identical cans with 250 cm^3 of hot water.
He wrapped strips of metal around them to model the elephants' ears.



He recorded the temperature of the water in each can every 5 minutes.
The table shows his results.

time (minutes)	temperature (°C)	
	can A	can B
0	60	60
5	54	57
10	50	54
15	46	52
20	43	50

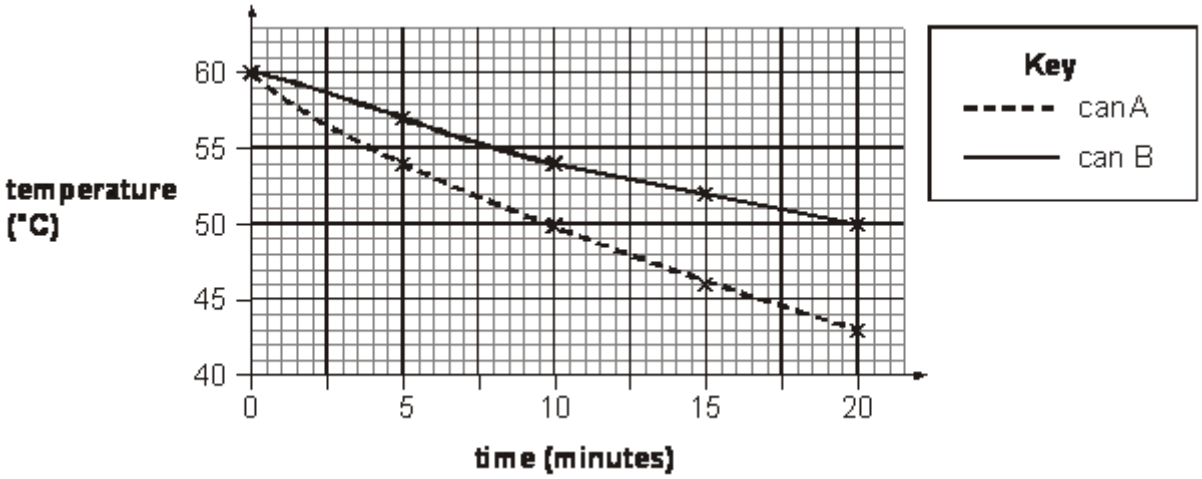
(i) Ben started with water at the same temperature in both cans.
Give **one** other way he made his test fair.

.....

.....

1 mark

(ii) He plotted the results for **can A** and **can B** and drew lines of best fit.



Why is it more useful to present these results in a graph rather than a table?

.....

.....

1 mark

(iii) The water in **can A** cooled more quickly than the water in **can B**.
Does this support your prediction in part (a)?
Tick the correct box.

yes ☐ no ☐

Explain your answer.

.....

.....

1 mark

(c) Ben repeated the investigation. Instead of a thermometer he used a temperature sensor and a data logger.
Give **one** advantage of this.

.....

.....

1 mark
maximum 5 marks

Q1.

(a) • African
its ears have a larger surface area

accept 'larger surface area'
accept 'larger ears'

***both** the answer and the correct explanation are required for the mark*
'they have large ears' is insufficient

1 (L5)

(b) (i) any **one** from

• he used identical cans

accept 'the cans were made of the same metal'
*accept 'the cans were the same size **or** shape'*
accept 'he used the same metal for the strips'

'he used metal for the strips' is insufficient

• he used the same amount of water

'he took the temperature every 5 minutes for both' is insufficient

accept 'he kept them in the same place'

1 (L5)

(ii) any **one** from

• the pattern **or** relationship can be seen more easily **or** more quickly

*accept 'the pattern **or** relationship is clearer'*

*'the pattern **or** the way the temperature changes can be seen' is insufficient*

accept 'the temperature change is clearer'

'it is clearer' is insufficient

accept 'the pattern can be seen straight away'

'to see them better' is insufficient

accept 'you can compare them more easily'

• anomalous results can be seen more quickly **or** easily

*accept 'you can make predictions more easily **or** more quickly'*

*do **not** accept 'a graph is more accurate'*

1 (L6)

(iii) • yes ✓
the greater the surface area of metal the greater the heat loss **or** the lower the temperature

if more than one box is ticked, award no mark
***both** the answer 'yes' and the correct explanation are required for the mark*

accept 'the bigger strips cause the greatest heat loss'

accept 'metal A represents big ears'

accept 'the can with bigger strips cooled more quickly'
*accept answers that relate the size **or** surface of the strips to the size **or** surface of the ears*

*do **not** accept 'can B was a better insulator'*

*accept 'no' and a relevant supporting explanation **only if** 'Indian elephant' is given as an answer to part (a). An example would be 'no, as I thought smaller ears would cool quicker'*

1 (L5)

(c) any **one** from

- it reduces error

  accept 'it is more accurate'
  *accept 'less human error' **or** 'is more reliable'*
  accept 'it is more precise'

- readings are continuous

  accept 'it takes more readings'

- it can draw a graph

  accept 'you can leave it to take results'

  'you can leave it' is insufficient

  accept 'it would record the results for him'

1 (L6)

[5]